

Data Converters

Exercise 05 – Noise Analysis of a Charge Redistribution Track & Hold Circuit

Shown in Figure 1 is a charge redistribution track & hold circuit. The equivalent circuit at the beginning of the sampling phase Φ_1 and redistribution phase Φ_2 is shown in Figure 2.

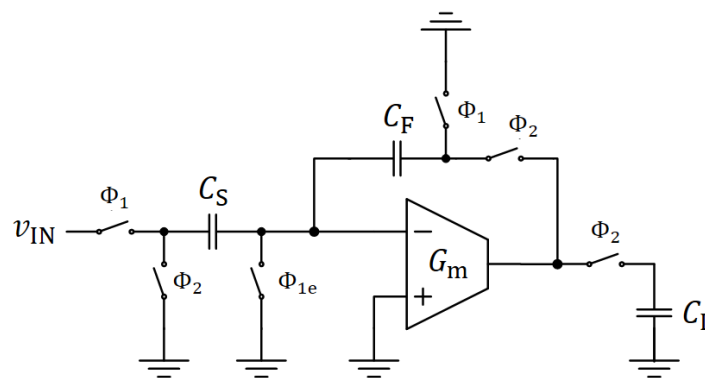


Figure 1: Charge redistribution track & hold circuit.

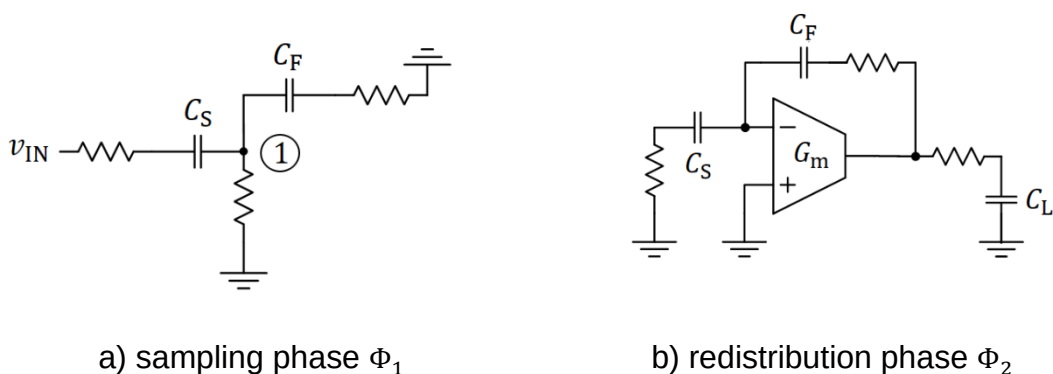


Figure 2: Equivalent circuit at the beginning of the sampling phase Φ_1 (a) and redistribution phase Φ_2 (b).

The settling behavior was analyzed in exercise 04 with the findings

$$\text{return factor } \beta = \frac{C_F}{C_F + C_S + C_X} \quad \text{output pole } p = \frac{\beta G_m}{C_L + (1 - \beta)C_F}$$

Considering the output pole p , it can be seen that the output resistance of the amplifier due to feedback results in

$$r_{\text{outF}} = \frac{1}{\beta G_m}$$

and the total capacitance at the output node

$$C_{L\text{tot}} = C_L + (1 - \beta)C_F$$

The task of this exercise is to analyze the noise behavior of the charge redistribution track & hold circuit.

Task I – Noise Analysis of the Sampling Phase

During the sampling phase, the on-resistances of the switches (see Figure 2.a) add kT/C -noise that is transferred to the output in the redistribution phase.

Straightforward yet cumbersome determination approach:

- Determine the transfer function of each noise source to node 1.
- Integrate the magnitude of the squared transfer functions from zero to infinity.
- Add the results.
- Transfer the resulting power to the output.

Better approach:

Apply the equipartition theorem which says

$$\frac{1}{2} \frac{\overline{q_1^2}}{C_{\text{eff}}} = \frac{1}{2} kT$$

i.e., that the total energy stored at node 1 equals $0.5kT$ in average.

- a. Refer to the paper *Thermal Noise in Track-and-Hold Circuits* by Boris Murmann and state the effective capacitor C_{eff} .
- b. Determine $\overline{q_1^2}$ based on your result of task I.a.
- c. Determine the transfer function of $\overline{q_1^2}$ to the output in the redistribution phase.
- d. How large is the output-referred noise power spectral density due to the sampling phase?

Task II – Noise Analysis of the Redistribution Phase

During the redistribution phase, the on-resistances of the switches (see Figure 2.b) add kT/C -noise to the noise power obtained from the sampling phase in task I. However, as will be discussed in the exercise, this noise power is negligible compared with the noise power of the G_m -cell.

The total output-referred noise of the G_m -cell is modelled by a current noise source according to Figure 3 with a power spectral density

$$\overline{i_n^2} = n_f \cdot 4kT\gamma G_m$$

The factor n_f accounts for the thermal noise contribution of several transistor to the overall output-referred noise power.

- Convert the current power spectral density $\overline{i_n^2}$ into the equivalent voltage power spectral density by means of the effective output resistance r_{outF} , which takes the effect of the feedback on the G_m -cell's resistance r_{out} into account.
- Since the feedback configuration is modelled by a single pole at the output, the total noise power of the G_m -cell can easily be calculated using the effective noise-bandwidth (NBF). Derive the NBF from the output pole p .
- Use the noise-bandwidth to determine the noise power of the G_m -cell.
- Determine the overall noise power due to the sampling and redistribution phase.

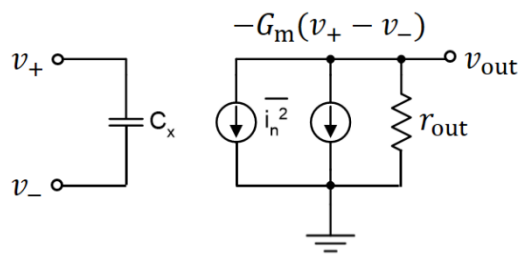


Figure 3: Small-signal equivalent circuit of the G_m -cell.

Task III – Numerical Example

This task summarizes the findings of PE IV and PE V by means of a numerical example.

The following parameters are given:

$$G_m = 2 \text{ mS} \quad g_{\text{out}} = 2 \text{ } \mu\text{S} \quad 20 \log_{10} G_m r_{\text{out}} = 60 \text{ dB} \quad \frac{g_m}{I_D} = \frac{G_m}{I_D} = 15 \frac{\text{S}}{\text{A}}$$

$$C_S = C_F = 0.1 \text{ pF} \quad C_L = 0.5 \text{ pF} \quad C_X = 0.021 \text{ pF}$$

$$\varepsilon_d = 0.2 \% \quad n_f = 2 \quad \gamma = 2/3$$

- Calculate the small-signal parameters feedback factor β , DC-gain A_{DC} , and the frequencies of the pole p and zero z of the closed-loop transfer function.
- Calculate the static settling error ε_s .
- For a differential signal $v_{\text{id}} = 10 \text{ mV}$ at the input of the G_m -cell, calculate the time it takes to reach a dynamic settling error $\varepsilon_d = 0.2 \%$.
- For a differential signal $v_{\text{id}} = 800 \text{ mV}$ at the input of the G_m -cell, calculate the time it takes to reach a dynamic settling error $\varepsilon_d = 0.2 \%$.
- Calculate the output-referred RMS-voltage $v_{\text{n,out,totalRMS}}$ of the output-referred noise power.